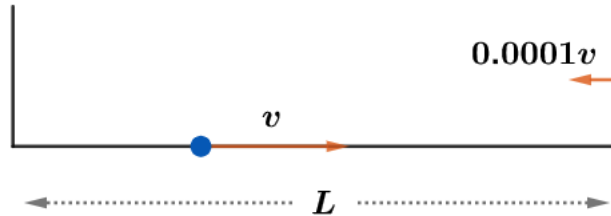


2021 F=ma Exam: Problem 7

Kevin S. Huang



The time for a round trip by the point mass is

$$t_0 = \frac{2L}{v}$$

while the time during which one of the walls moves is

$$t = \frac{0.01L}{0.0001v} = \frac{100L}{v}$$

so the mass has $N = t/t_0 = 50$ collisions with the moving wall.



For an elastic collision with a wall moving at speed u , the particle picks up an additional $2u$ in speed (seen by going to wall frame). Applying this to our case,

$$\Delta v = N(2u) = 0.01v$$

$$v_f = v + \Delta v = 1.01v$$

so the answer is D.